

Data Science Fundamentals [SC3021]

Chapter 4: Data Preparation

Assoc. Prof. Melanie Herschel | CCDS

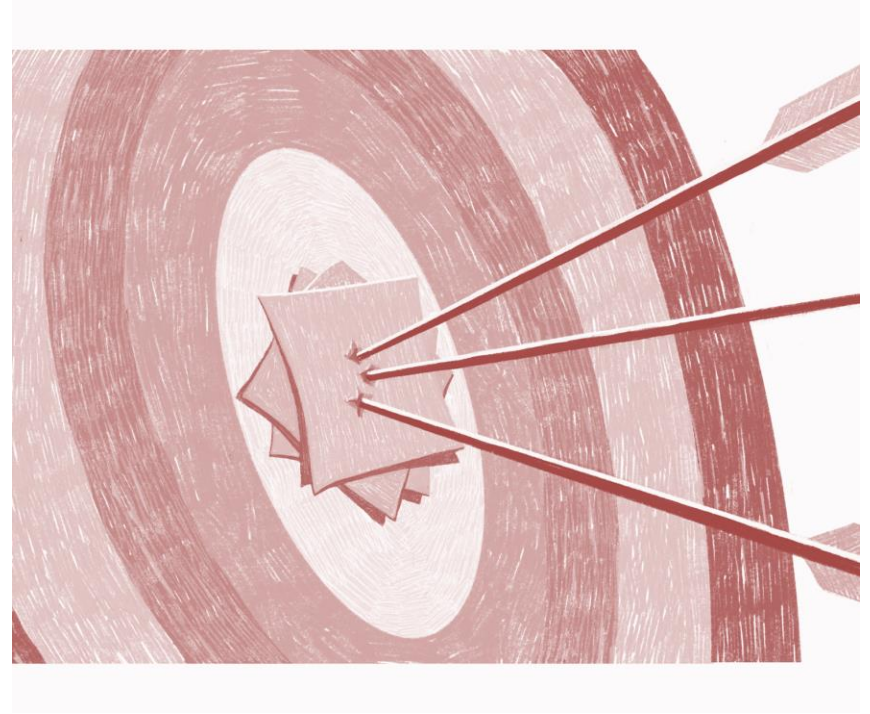


**NANYANG
TECHNOLOGICAL
UNIVERSITY**
SINGAPORE

Chapter 4.2: Data profiling

At the end of this sub-chapter, you should be able to...

- ... explain different profiling tasks and how they may apply at different data science stages.
- ... identify profiling tasks suited to gain relevant information on a given dataset for a specific application.
- ... reflect on the computational complexity of profiling tasks when applying them to large data sets.



INTRODUCTION TO DATA PROFILING

Data profiling in the Data Science Ecosystem



Data Engineering

The practice of designing, building, and maintaining systems that collect, manage, and convert raw data into usable information for downstream applications.

Data preparation

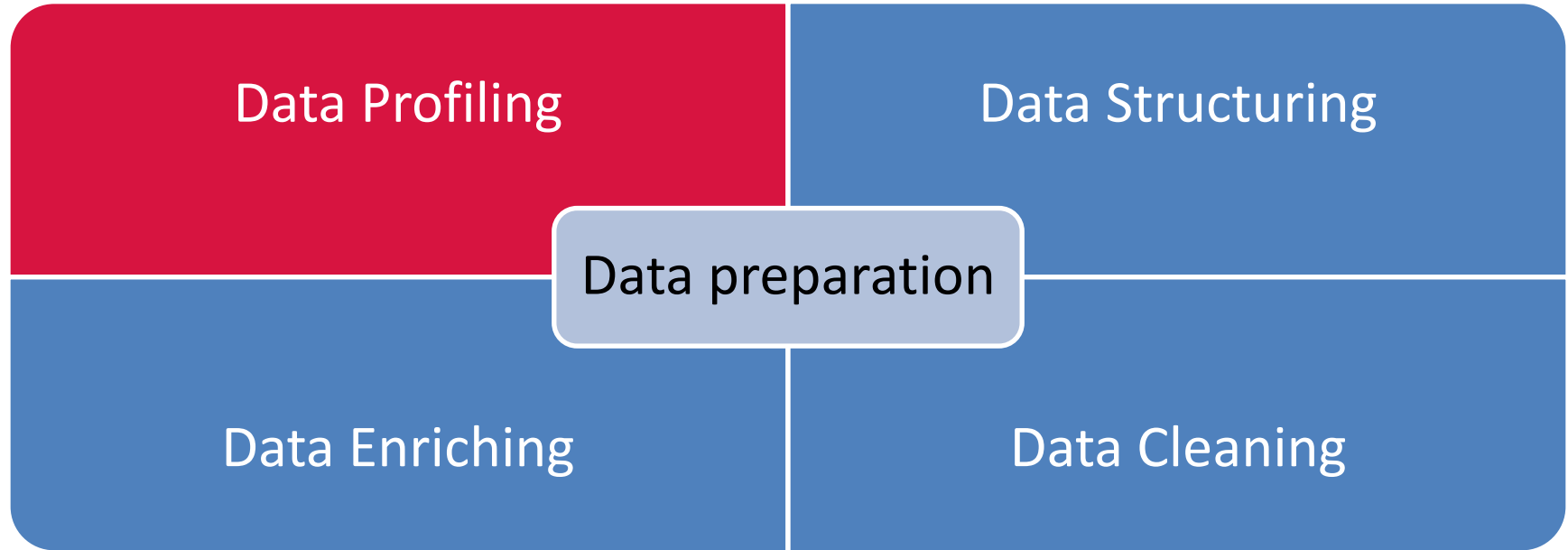
Operations typically involved in a data engineering process that transform the data.

Data profiling fits here

Data management

Solutions devised to organize, protect, store, access, and manipulate data.

Data preparation tasks



Importance of data profiling

- Ensure ingested data is “as expected”
- Understand main data characteristics
- Assess data utility using data characteristics
- Inspect data to validate and refine data transformation steps.
- Monitor essential data characteristics.

raw

refined

production

DATA PROFILING TASKS

Intuitive data profiling introduction (reminder)

- Examine and analyze data to create metadata (data about data).
- Metadata represent a summary of various characteristics of the data.
- These summaries help to assess usefulness and quality of data.

Simple examples:

- Number of rows
- Min and max values
- Number of missing values
- Outliers
- Data type
- Data format violations
- Histograms
- ...

Downloaded file about registered voters

File Edit View

ncvoter1.txt

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60	61	62	63	64	65	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85	86	87	88	89	90	91	92	93	94	95	96	97	98	99	100	101	102	103	104	105	106	107	108	109	110	111	112	113	114	115	116	117	118	119	120	121	122	123	124	125	126	127	128	129	130	131	132	133	134	135	136	137	138	139	140	141	142	143	144	145	146	147	148	149	150	151	152	153	154	155	156	157	158	159	160	161	162	163	164	165	166	167	168	169	170	171	172	173	174	175	176	177	178	179	180	181	182	183	184	185	186	187	188	189	190	191	192	193	194	195	196	197	198	199	200	201	202	203	204	205	206	207	208	209	210	211	212	213	214	215	216	217	218	219	220	221	222	223	224	225	226	227	228	229	230	231	232	233	234	235	236	237	238	239	240	241	242	243	244	245	246	247	248	249	250	251	252	253	254	255	256	257	258	259	260	261	262	263	264	265	266	267	268	269	270	271	272	273	274	275	276	277	278	279	280	281	282	283	284	285	286	287	288	289	290	291	292	293	294	295	296	297	298	299	300	301	302	303	304	305	306	307	308	309	310	311	312	313	314	315	316	317	318	319	320	321	322	323	324	325	326	327	328	329	330	331	332	333	334	335	336	337	338	339	340	341	342	343	344	345	346	347	348	349	350	351	352	353	354	355	356	357	358	359	360	361	362	363	364	365	366	367	368	369	370	371	372	373	374	375	376	377	378	379	380	381	382	383	384	385	386	387	388	389	390	391	392	393	394	395	396	397	398	399	400	401	402	403	404	405	406	407	408	409	410	411	412	413	414	415	416	417	418	419	420	421	422	423	424	425	426	427	428	429	430	431	432	433	434	435	436	437	438	439	440	441	442	443	444	445	446	447	448	449	450	451	452	453	454	455	456	457	458	459	460	461	462	463	464	465	466	467	468	469	470	471	472	473	474	475	476	477	478	479	480	481	482	483	484	485	486	487	488	489	490	491	492	493	494	495	496	497	498	499	500	501	502	503	504	505	506	507	508	509	510	511	512	513	514	515	516	517	518	519	520	521	522	523	524	525	526	527	528	529	530	531	532	533	534	535	536	537	538	539	540	541	542	543	544	545	546	547	548	549	550	551	552	553	554	555	556	557	558	559	560	561	562	563	564	565	566	567	568	569	570	571	572	573	574	575	576	577	578	579	580	581	582	583	584	585	586	587	588	589	590	591	592	593	594	595	596	597	598	599	600	601	602	603	604	605	606	607	608	609	610	611	612	613	614	615	616	617	618	619	620	621	622	623	624	625	626	627	628	629	630	631	632	633	634	635	636	637	638	639	640	641	642	643	644	645	646	647	648	649	650	651	652	653	654	655	656	657	658	659	660	661	662	663	664	665	666	667	668	669	670	671	672	673	674	675	676	677	678	679	680	681	682	683	684	685	686	687	688	689	690	691	692	693	694	695	696	697	698	699	700	701	702	703	704	705	706	707	708	709	710	711	712	713	714	715	716	717	718	719	720	721	722	723	724	725	726	727	728	729	730	731	732	733	734	735	736	737	738	739	740	741	742	743	744	745	746	747	748	749	750	751	752	753	754	755	756	757	758	759	760	761	762	763	764	765	766	767	768	769	770	771	772	773	774	775	776	777	778	779	780	781	782	783	784	785	786	787	788	789	790	791	792	793	794	795	796	797	798	799	800	801	802	803	804	805	806	807	808	809	810	811	812	813	814	815	816	817	818	819	820	821	822	823	824	825	826	827	828	829	830	831	832	833	834	835	836	837	838	839	840	841	842	843	844	845	846	847	848	849	850	851	852	853	854	855	856	857	858	859	860	861	862	863	864	865	866	867	868	869	870	871	872	873	874	875	876	877	878	879	880	881	882	883	884	885	886	887	888	889	890	891	892	893	894	895	896	897	898	899	900	901	902	903	904	905	906	907	908	909	910	911	912	913	914	915	916	917	918	919	920	921	922	923	924	925	926	927	928	929	930	931	932	933	934	935	936	937	938	939	940	941	942	943	944	945	946	947	948	949	950	951	952	953	954	955	956	957	958	959	960	961	962	963	964	965	966	967	968	969	970	971	972	973	974	975	976	977	978	979	980	981	982	983	984	985	986	987	988	989	990	991	992	993	994	995	996	997	998	999	1000	1001	1002	1003	1004	1005	1006	1007	1008	1009	1010	1011	1012	1013	1014	1015	1016	1017	1018	1019	1020	1021	1022	1023	1024	1025	1026	1027	1028	1029	1030	1031	1032	1033	1034	1035	1036	1037	1038	1039	1040	1041	1042	1043	1044	1045	1046	1047	1048	1049	1050	1051	1052	1053	1054	1055	1056	1057	1058	1059	1060	1061	1062	1063	1064	1065	1066	1067	1068	1069	1070	1071	1072	1073	1074	1075	1076	1077	1078	1079	1080	1081	1082	1083	1084	1085	1086	1087	1088	1089	1090	1091	1092	1093	1094	1095	1096	1097	1098	1099	1100	1101	1102	1103	1104	1105	1106	1107	1108	1109	1110	1111	1112	1113	1114	1115	1116	1117	1118	1119	1120	1121	1122	1123	1124	1125	1126	1127	1128	1129	1130	1131	1132	1133	1134	1135	1136	1137	1138	1139	1140	1141	1142	1143	1144	1145	1146	1147	1148	1149	1150	1151	1152	1153	1154	1155	1156	1157	1158	1159	1160	1161	1162	1163	1164	1165	1166	1167	1168	1169	1170	1171	1172	1173	1174	1175	1176	1177	1178	1179	1180	1181	1182	1183	1184	1185	1186	1187	1188	1189	1190	1191	1192	1193	1194	1195	1196	1197	1198	1199	1200	1201	1202	1203	1204	1205	1206	1207	1208	1209	1210	1211	1212	1213	1214	1215	1216	1217	1218	1219	1220	1221	1222	1223	1224	1225	1226	1227	1228	1229	1230	1231	1232	1233	1234	1235	1236	1237	1238	1239	1240	1241	1242	1243	1244	1245	1246	1247	1248	1249	1250	1251	1252	1253	1254	1255	1256	1257	1258	1259	1260	1261	1262	1263	1264	1265	1266	1267	1268	1269	1270	1271	1272	1273	1274	1275	1276	1277	1278	1279	1280	1281	1282	1283	1284	1285	1286	1287	1288	1289	1290	1291	1292	1293	1294	1295	1296	1297	1298	1299	1300	1301	1302	1303	1304	1305	1306	1307	1308	1309	1310	1311	1312	1313	1314	1315	1316	1317	1318	1319	1320	1321	1322	1323	1324	1325	1326	1327	1328	1329	1330	1331	1332	1333	1334	1335	1336	1337	1338	1339	1340	1341	1342	1343	1344	1345	1346	1347	1348	1349	1350	1351	1352	1353	1354	1355	1356	1357	1358	1359	1360	1361	1362	1363	1364	1365	1366	1367	1368	1369	1370	1371	1372	1373	1374	1375	1376	1377	1378	1379	1380	1381	1382	1383	1384	1385	1386	1387	1388	1389	1390	1391	1392	1393	1394	1395	1396	1397	1398	1399	1400	1401	1402	1403	1404	1405	1406	1407	1408	1409	1410	1411	1412	1413	1414	1415	1416	1417	1418	1419	1420	1421	1422	1423	1424	1425	1426	1427	1428	1429	1430	1431	1432	1433	1434	1435	1436	1437	1438	1439	1440	1441	1442	1443	1444	1445	1446	1447	1448	1449	1450	1451	1452	1453	1454	1455	1456	1457	1458	1459	1460	1461	1462	1463	1464	1465	1466	1467	1468	1469	1470	1471	1472	1473	1474	1475	1476	1477	1478	1479	1480	1481	1482	1483	1484	1485	1486	1487	1488	1489
---	---	---	---	---	---	---	---	---	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------

Manual profiling example using Excel

Column headers

	X	Y	Z	AA	AB	AC	AD	AE	AF	AG	AH	AI	AJ		
1	code	full_phone_number	confidential_ind	registr_dt	race_code	ethnic_code	party_cd	gender_code	birth_year	age_at_year_end	birth_state	drivers_lic	precinct_abbrv	precinct_desc	municip
3			N	02/23/2018	W	UN	UNA	F	1978		46 DC	Y			
4	27302		N	10/31/2020	W	UN	REP	M	1966		58 AL	N	103	MELVILLE 3	
5	27215	3362291110	N	03/26/1996	W	UN	UNA	F	1976		48 NC	Y	03SE	SOUTH BOONE E	BUR
6	27215	2228834	N	08/15/1989	W	NL	UNA	F	1945		79 VA	Y	124	BURLINGTON 4	BUR
7	27340	2027443411	N	03/07/2012	W	UN	DEM	M	1948		76 MA	N	08N	NORTH NEWLIN	
8	27215	2529408167	N	06/01/2020	B	NL	DEM	F	1966		58 NC	Y	03SE	SOUTH BOONE E	BUR
9	27215	3362291110	N	10/10/1994	W	UN	UNA	M	1976		48 NC	Y	03SE	SOUTH BOONE E	BUR
10	27215	3362228834	N	12/05/2013	W	UN	UNA	M	1972		52 VA	Y	124	BURLINGTON 4	BUR
11	27215	3362228834	N	12/05/2013	W	HL	UNA	F	1975		49	N	124	BURLINGTON 4	BUR
12	27215	3362228834	N	06/06/1990	W	NL	REP	M	1944		80 VA	Y	124	BURLINGTON 4	BUR
13	27258	3365789123	N	08/18/1998	W	NL	REP	F	1972		52 NC	Y	13	HAW RIVER	HAW
14	27258	3362667615	N	01/19/2006	W	NL	REP	M	1962		62 WI	Y	13	HAW RIVER	HAW
15	27253		N	05/06/2020	W	NL	UNA	M	1987		37	Y	06N	NORTH GRAHAM	GRA
16	27253	3362644972	N	01/16/2009	W	NL	REP	M	1986		38 NC	Y	06N	NORTH GRAHAM	GRA
17	27253	3366757112	N	06/12/2017	W	UN	UNA	M	1960		64 LA	Y	06N	NORTH GRAHAM	GRA
18	27253	3365127765	N	12/02/2008	W	UN	REP	F	1951		73 NC	Y	06N	NORTH GRAHAM	GRA
19	27258		N	08/10/2018	U	UN	REP	U	1959		65	Y	09N	NORTH THOMPSON	
20	27253	8133102051	N	06/07/2024	W	UN	REP	F	1950		74 NY	Y	09N	NORTH THOMPSON	SWE
21	27302	9196327914	N	11/01/2024	W	NL	UNA	F	1970		54	Y	10N	NORTH MELVILLE	MEB
22	27253	8133100652	N	05/09/2023	W	NL	REP	M	1945		79 NY	Y	09N	NORTH THOMPSON	SWE
23	27302	7049961866	N	08/26/2016	W	NL	UNA	F	1990		34 NC	Y	10S1	SOUTH MELVILLE	MEB
24	27302		N	09/08/2016	W	NL	UNA	M	1989		35 NC	Y	10S1	SOUTH MELVILLE	MEB
25	27244	9147724005	N	10/04/2024	W	NL	DEM	U	2003		21	N	03N	NORTH BOONE	ELO
26	27258	4302953995	N	05/17/2023	W	NL	REP	F	1967		57 FL	N	13	HAW RIVER	
27	27215	3365850666	N	10/26/2012	W	NL	DEM	F	1980		44 FL	Y	03C	CENTRAL BOONE	BUR
28			N	03/30/2000	A	UN	DEM	F	1952		72 NC	Y			
29	27244	4018353485	N	10/11/2024	W	NL	REP	U	2004		20	N	035	BOONE 5	ELO
30	27249	3366842638	N	04/15/2021	W	NL	REP	M	2003		21 NC	Y	03W	WEST BOONE	GIB
31	27249	3363398364	N	12/09/2019	W	NL	REP	F	2002		22 NC	Y	03W	WEST BOONE	GIB
32	27249	3364499029	N	05/14/1996	W	UN	UNA	M	1976		48 NY	Y	03W	WEST BOONE	GIB
33	27249		N	10/09/2024	W	NL	UNA	U	1979		45	Y	03W	WEST BOONE	GIB

Manual profiling example using Excel

	county_id	county_desc	voter_reg_num	ncid	last_name	first_name	middle_name	name_suffix_lbl	status_cd	voter_status_desc	reason_cd	voter_status_reason
135487	1	ALAMANCE	9194677	AA211067	ZUNIGA CAMBERO	ANAKAREN			A	ACTIVE	AV	VERIFIED
135488	1	ALAMANCE	9236267	AA234180	ZUNIGA CAMBERO	NORA	ELIA		A	ACTIVE	AV	VERIFIED
135489	1	ALAMANCE	9201310	AA214486	ZUNIGA FIERRO	CHRISTIAN	DENNIS		A	ACTIVE	AV	VERIFIED
135490	1	ALAMANCE	9206554	BL513369	ZUNIGA FIERRO	JISSEL			A	ACTIVE	AV	VERIFIED
135491	1	ALAMANCE	9200517	AA214045	ZUNIGA HERRERA	MIGUEL			A	ACTIVE	AV	VERIFIED
135492	1	ALAMANCE	9141392	AA179328	ZUPANCICH	MONICA	ANITA		I	INACTIVE	IN	CONFIRMATION NO
135493	1	ALAMANCE	9141406	AA145791	ZUPANCICH	RONALD	JAMES	II	I	INACTIVE	IN	CONFIRMATION NO
135494	1	ALAMANCE	9152399	AA186234	ZURIFF	SOPHIE	ANNA		I	INACTIVE	IU	CONFIRMATION RET
135495	1	ALAMANCE	9166564	AA194726	ZWELLING	AMY	MARIE		A	ACTIVE	AV	VERIFIED
135496	1	ALAMANCE	9230103	AA230908	ZWELLING	JOEDIN	SHEA		A	ACTIVE	AV	VERIFIED
135497	1	ALAMANCE	9187816	AA206903	ZWELLING	PAUL	J		A	ACTIVE	AV	VERIFIED
135498	1	ALAMANCE	9174660	BN352594	ZWICK	MICHAEL	TODD		A	ACTIVE	AV	VERIFIED
135499	1	ALAMANCE	9099472	AA148089	ZWIER	ANDREW	MICHAEL		A	ACTIVE	AV	VERIFIED
135500	1	ALAMANCE	9099260	AA147942	ZWIER	CHRISTOPHER	ANTHONY		A	ACTIVE	AV	VERIFIED
135501	1	ALAMANCE	9228255	DK27494	ZWIER	JESSICA			A	ACTIVE	AV	VERIFIED
135502	1	ALAMANCE	9099261	AA147943	ZWIER	KAREN	JEAN		A	ACTIVE	AV	VERIFIED
135503	1	ALAMANCE	9217806	DE289078	ZWILLING	JASON	ERIC		A	ACTIVE	AV	VERIFIED
135504	1	ALAMANCE	9146897	AA182983	ZWINGELBERG	ANNA	MARIE		D	DENIED	DU	VERIFICATION RETU
135505	1	ALAMANCE	9176512	BL455029	ZYCZKIEWICZ	DIANE			A	ACTIVE	AV	VERIFIED
135506	1	ALAMANCE	9196290	DE247523	ZYCZKIEWICZ	KRISTINE	MARIE		A	ACTIVE	AV	VERIFIED
135507	1	ALAMANCE	9196289	AX71581	ZYCZKIEWICZ	RICHARD			A	ACTIVE	AV	VERIFIED
135508	1	ALAMANCE	9236502	AA234296	ZYLOWSKI	PATRICIA	ANN		A	ACTIVE	AV	VERIFIED
135509	1	ALAMANCE	9161152	AA191482	ZYMECK	JULIET	LORE		I	INACTIVE	IU	CONFIRMATION RET
135510												
135511												
135512												
135513												
135514												
135515												
135516												
135517												
135518												
135519												



Number of rows

Manual profiling example using Excel

	A	B	C	D	E	F	G	H	I	J	K	L
	county_id	county_desc	voter_reg_num	ncid	last_name	first_name	middle_name	name_suffix_lbl	status_cd	voter_status_desc	reason_cd	voter_status_reason_d
2	1	ALAMANCE	9005990	AA56273	AABEL	RUTH	EVELYN			REMOVED	RD	DECEASED
3	1	ALAMANCE	9178574	AA201627	AARDEN	JONI	AUTUMN			REMOVED	RL	MOVED FROM COUNTY
4	1	ALAMANCE	9205561	AA216996	AARMSTRONG	TIMOTHY	DUANE			ACTIVE	AV	VERIFIED
5	1	ALAMANCE	9048723	AA98377	AARON	CHRISTINA	CASTAGNA			ACTIVE	AV	VERIFIED
6	1	ALAMANCE	9019674	AA69747	AARON	CLAUDIA	HAYDEN			ACTIVE	AV	VERIFIED
7	1	ALAMANCE	9129589	AA170513	AARON	JAMES	MICHAEL			ACTIVE	AV	VERIFIED
8	1	ALAMANCE	9194595	BM49306	AARON	KIMBERLY	GREEN			INACTIVE	IU	CONFIRMATION RETUR
9	1	ALAMANCE	9041748	AA91549	AARON	NATHAN	EDWARD			ACTIVE	AV	VERIFIED
10	1	ALAMANCE	9144384	AA125250	AARON	RICHARD	BRIAN			ACTIVE	AV	VERIFIED
11	1	ALAMANCE	9144385	AA181361	AARON	SANDRA	ESCOBAR			ACTIVE	AV	VERIFIED
12	1	ALAMANCE	9021947	AA71983	AARON	WILLIE	DALE			ACTIVE	AV	VERIFIED
13	1	ALAMANCE	9062002	AA111384	AARONSON	GENA	HOLT			ACTIVE	AV	VERIFIED
14	1	ALAMANCE	9096423	AA145641	AARONSON	MICHAEL	CHARLES			ACTIVE	AV	VERIFIED
15	1	ALAMANCE	9194267	AA210859	AASEN	ALEXANDER	MACNAB			INACTIVE	IN	CONFIRMATION NOT R
16	1	ALAMANCE	9121656	AA165057	ABADIE	JACK	EDWARD			ACTIVE	AV	VERIFIED
17	1	ALAMANCE	9168958	AA196147	ABADIE	JACK	EDWARD			ACTIVE	AV	VERIFIED
18	1	ALAMANCE	9118154	AA162459	ABADIE	MYRA	LYNN			ACTIVE	AV	VERIFIED
19	1	ALAMANCE	9175986	AA200312	ABADIE	WILLIAM	EARL			ACTIVE	AP	VERIFICATION PENDING
20	1	ALAMANCE	9233812	AA233007	ABAJIAN	DEBORAH	RUTH			ACTIVE	AV	VERIFIED
21	1	ALAMANCE	9242743	BL377881	ABAJIAN	MICHELLE	S			ACTIVE	AV	VERIFIED
22	1	ALAMANCE	9225186	AA228074	ABAJIAN	ROBERT	HAROLD			ACTIVE	AV	VERIFIED
23	1	ALAMANCE	9160433	CW803328	ABASHIAN	ASHLEY	DENNY PARR			ACTIVE	AV	VERIFIED
24	1	ALAMANCE	9160854	EH730361	ABASHIAN	MICHAEL	DAVID			ACTIVE	AV	VERIFIED
25	1	ALAMANCE	9239089	AA235710	ABATEMARCO	MEGAN	REID			ACTIVE	AV	VERIFIED
26	1	ALAMANCE	9225335	AA228157	ABAZIED COOPER	JOY	GAYLE			ACTIVE	AV	VERIFIED
27	1	ALAMANCE	9139919	AA178336	ABBAS	JENNIFER	ANN		A	ACTIVE	AV	VERIFIED
28	1	ALAMANCE	9068460	AA117728	ABBAS	RAFAT			R	REMOVED	RS	MOVED FROM STATE
29	1	ALAMANCE	9239920	AA236185	ABBATE	MICHAEL	EDWIN		A	ACTIVE	AV	VERIFIED
30	1	ALAMANCE	9209221	AA219101	ABBATECOLA	ANTHONY	JOSEPH		A	ACTIVE	AV	VERIFIED
31	1	ALAMANCE	9189448	AA207788	ABBATECOLA	JENNA	CAMILLE		A	ACTIVE	AV	VERIFIED
32	1	ALAMANCE	9049573	AA99206	ABBATECOLA	RONALD	JOSEPH	JR	A	ACTIVE	AV	VERIFIED
33	1	ALAMANCE	9240574	BY754441	ABBATECOLA	THOMAS	PAUL		A	ACTIVE	AV	VERIFIED

Sort A to Z
Sort Z to A
Sort by Color
Sheet View
Clear Filter From "status_cd"
Filter by Color
Text Filters

Search

☒ (Select All)

☒ A
☒ D
☒ I
☒ R
☒ S

OK Cancel

Distinct values

Manual profiling example using Excel

U V W X Y Z AA AB AC AD AE AF AG

1 mail_city mail_state mail_zipcode full_phone_number confidential_ind registr_dt race_code ethnic_code party_cd gender_code birth_year age_at_year_end birth_state

1928 MEBANE NC 27302 33 W NL UNA F 1974 50 NC

3560 W NL REP F 1932 92 NC

5192 BURLINGTON NC 27216 33 W UN REP M 1949 75 NC

6176 BURLINGTON NC 27215 33 W NL DEM F 1929 95 NC

6305 SNOW CAMP NC 27349 33 W NL REP M 1971 53 NC

7029 W NL DEM M 1938 86 NC

7107 GRAHAM NC 27253 33 W NL UNA M 1941 83 NC

8376 BURLINGTON NC 27217 33 W NL REP M 1984 40 VA

11570 W NL DEM F 1930 94 NC

12322 BURLINGTON NC 27217 33 B NL DEM F 1947 77 NC

12354 GRAHAM NC 27253 33 W NL REP F 1933 91 NC

12855 SNOW CAMP NC 27349 33 W NL UNA F 1941 83 NC

13039 GRAHAM NC 27253 33 B UN DEM M 1977 47 NC

14373 BURLINGTON NC 27215 33 W NL UNA F 1932 92 SC

14589 W NL DEM M 1943 81 NC

19633 BURLINGTON NC 27217 33 W NL REP F 1937 87 NC

20024 BURLINGTON NC 27217 33 W NL DEM M 1942 82 NC

20072 HAW RIVER NC 27217 33 W NL DEM M 1944 80 NC

21780 W NL DEM M 1928 96 PA

22026 GRAHAM NC 27253 33 B NL DEM F 2002 22 VA

22936 W NL DEM F 1929 95 NC

23877 BURLINGTON NC 27215 33 W NL REP F 1938 86 NC

24141 ELON NC 27244 33 W NL REP M 1939 85 NC

24886 W NL UNA F 1939 85 NC

25994 MEBANE NC 27349 33 W NL REP M 1945 79 DC

27094 SNOW CAMP NC 27349 33 W NL REP M 1948 76 NC

27755 N ##### W NL REP F 1932 92 NC

27759 N ##### W UN REP M 1929 95 NC

29087 N ##### W NL REP F 1928 96 OC

29725 BURLINGTON NC 27215 3362145254 N ##### W NL REP F 1931 93 NC

29731 BURLINGTON NC 27215 0 N ##### W NL REP M 1930 94 NC

33457 N ##### W NL REP F 1939 85 NC

Sort A to Z
Sort Z to A
Sort by Color
Sheet View
Clear Filter From "registr_dt"
Filter by Color
Text Filters
Search
(Select All)
[x] ###/###/####
[] 01/01/1976
[] 01/01/2019
[] 01/01/2020
[] 01/01/2021
[] 01/01/2022
[] 01/01/2023
[] 01/01/2024
Not all items showing
OK Cancel

Number of dummy values

Number of dummy values

Ready 135 of 135508 records found Accessibility: Investigate

Count: 136 Display Settings

Manual profiling example using Excel

	thnic_code	party_cd	gender_code	birth_year	age_at_year_end	birth_state	drivers_lic	precinct_abbrv	precinct_desc	municipality_abbrv	municipality_desc	ward_abbrv	ward_desc	cong_dist_abbr
4	IN	REP	M	1966		58 AL	N	103	M					
7	IN	DEM	M	1948		76 MA	N	08N	N					
19	IN	REP	U	1959		65	Y	09N	N					
26	IL	REP	F	1967		57 FL	N	13	H					
28	IN	DEM	F	1952		72 NC	Y							
35	IL	REP	M	1968		56 NY	Y	07	AI					
36	IL	REP	F	1968		56 SC	Y	07	AI					
37	IL	REP	M	2000		24 NC	N	07	AI					
40	IL	UNA	F	1996		28 NJ	N							
44	IL	UNA	M	1968		56 OH	Y							
45	IL	DEM	M	1955		69 NC	Y	127	B					
46	IN	REP	M	1959		65 NC	Y	09S	S					
48	IL	DEM	F	1952		72 NC	Y							
51	IL	UNA	F	1983		41 NC	Y	09S	S					
54	IL	DEM	F	1985		39 NC	Y							
56	IL	REP	M	1978		46 NC	Y	02	C					
60	IL	DEM	F	2001		23 PA	N							
61	IL	REP	F	1992		32 VA	Y	10N	N					
62	IL	DEM	F	1953		71	Y	103	M					
63	IN	UNA	U	1965		59	Y							
65	IL	DEM	F	1973		51 NC	Y	127	B					
86	IL	UNA	F	1991		33 NC	Y	10N	N					
87	IL	UNA	M	1962		62 NC	Y		B					
88	IL	DEM	F	1959		65 CA	Y		AI					
99	IL	UNA	M	1964		60 NY		08S	S					
101	IL	UNA	F	1998		26 NC	Y	09S	S					
102	IN	DEM	U	1990		34	Y	09S	S					
103	IL	DEM	F	1966		39		09S	S					
105	IL	REP	M	1992		32 NC	Y	035	S					
106	IL	DEM	M	1965		59 NC	Y	09S	S					
107	IL	REP	M	1996		28 NC	Y	09S	S					
108	IL	REP		1989		35 NC	Y	035	B					
110	IN			1981		43	Y	06S	S					

Number of missing values

Sort A to Z
Sort Z to A
Sort by Color
Sheet View
Clear Filter From "municipality_abbrv"
Filter by Color
Text Filters
Search
☐ ELO
☐ GIB
☐ GRA
☐ GRE
☐ HAW
☐ MEB
☐ OSS
☐ SWE
☒ (Blanks)
OK Cancel

Manual profiling example using Excel

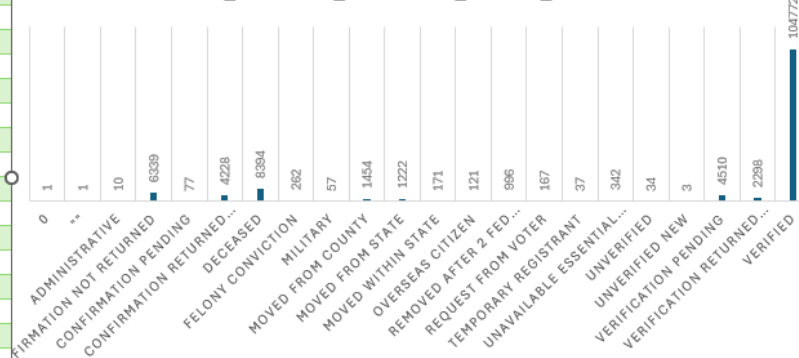
	X	Y	Z	AA	AB	AC	AD	AE	AF	AG	AH	AI	AJ	AK
1	full_phone_number	confidential_ind	registr_dt	race_code	ethnic_code	party_cd	gender_code	birth_year	age_at_year_end	birth_state	drivers_lic	precinct_abbrv	precinct_desc	municipality_ab
2	3362277017	N	11/02/2004	W	UN	REP	F	1909	115	NC	N			
3		N	05/11/1946	B	NL	DEM	F	1911	113	NC	Y			
4		N	10/02/2008	B	UN	DEM	F	1911	113	NC	N			
5		N	10/28/1950	W	NL	REP	F	1911	113	NC	Y			
6		N	10/06/1986	B	NL	DEM	F	1913	111	NC	N			
7		N	10/26/1956	U	NL	DEM	M	1914	110	NC	Y			
8		N	06/24/1994	W	NL	DEM	F	1915	109	NC	Y			
9		N	04/20/1968	W	NL	REP		1915	109	NC	Y			
10		N	11/08/2016	U	UN	DEM		1915	109	NC	Y	10S1	SOUTH MELVILLE	
11		N	10/06/2016	U	UN	REP		1916	108	MA	N			
12		N	09/21/2016	W	UN		F	1916	108	VA	N			
13		N	05/14/1960	B	NL		F	1916	108	NC	N			
14		N	06/20/2000	W	NL		M	1916	108	IN	Y			
15		N	09/13/2016	W	NL		F	1916	108	PA	Y			
16		N	05/17/1952	W	NL	DEM	F	1916	108	NC	Y			
Min/max values														
135494	3363802385	N	05/28/2024	W	UN	UNA	M	2006	18	NY	Y	11	PLEASANT GROVE	
135495	3362636795	N	06/13/2024	U	HL	UNA	F	2006	18		Y	064	GRAHAM 4	GRA
135496	5512327930	N	01/05/2024	B	UN	UNA	M	2006	18	PA	Y	1210	BURLINGTON 10	BUR
135497		N	09/13/2024	W	NL		F	2006	18		Y	03N	NORTH BOONE	ELO
135498	3365163246	N	01/05/2024	W	NL		M	2006	18	NC	Y	07	ALBRIGHT	
135499	3362601958	N	08/29/2024	W	UN	UNA	M	2006	18	NC	Y	10S1	SOUTH MELVILLE	
135500	9193049069	N	01/05/2024	W	NL	REP		2006	18	NC	Y			
135501		N	08/25/2024	W	NL	REP		2006	18		Y	10S1	SOUTH MELVILLE	
135502	3362662530	N	01/05/2024	W	UN	UNA		2006	18	NC	Y	03W	WEST BOONE	
135503	3365241260	N	01/05/2024	W	NL	REP	F	2006	18	NC	Y	05	FAUCETTE	
135504	3362788618	N	01/05/2024	W	NL	REP	M	2006	18	NC	Y	08S	SOUTH NEWLIN	
135505	3365346144	N	01/05/2024	W	HL	UNA	F	2006	18	WI	Y	03SM	SOUTH BOONE M	BUR
135506	3363435524	N	01/05/2024	O	UN	DEM	F	2006	18	NC	Y	06E	EAST GRAHAM	GRA
135507	3362703926	N	09/12/2024	W	UN	UNA	F	2006	18	NC	Y	125	BURLINGTON 5	BUR
135508	3365679723	N	01/05/2024	W	NL	UNA	M	2006	18	MI	Y	13	HAW RIVER	
135509	6153103615	N	01/05/2024	W	NL	REP	F	2006	18	NY	Y	11	PLEASANT GROVE	
135510														
135511														

Manual profiling example using Excel

	first_name	middle_name	name_suffix_lbl	status_cd	voter_status_desc	reason_cd	voter_status_reason_desc	res_street_address	res_city_desc	state_cd
2	RUTH	EVELYN		R	REMOVED	RD	DECEASED	REMOVED		
3	JONI	AUTUMN		R	REMOVED	RL	MOVED FROM COUNTY	REMOVED		
4	TIMOTHY	DUANE		A	ACTIVE	AV	VERIFIED	3670 COVINGTON TRL	MEBANE	NC
5	WILLIE	DALE		A	ACTIVE	AV	VERIFIED	1013 EDITH ST	BURLINGTON	NC
6	CLAUDIA	HAYDEN		A	ACTIVE	AV	VERIFIED	1013 EDITH ST	BURLINGTON	NC
7	JAMES	MICHAEL		A	ACTIVE	AV	VERIFIED	5608 OLD CHEROKEE LN	GRAHAM	NC
8	KIMBERLY	GREEN		I	INACTIVE	IU	CONFIRMATION RETURNED UNDELIVERABLE	187 CAVALIER WAY #104	BURLINGTON	NC
9	RICHARD	BRIAN		A	ACTIVE	AV	VERIFIED	1013 EDITH ST	BURLINGTON	NC
10	SANDRA	ESCOBAR		A	ACTIVE	AV	VERIFIED	1013 EDITH ST	BURLINGTON	NC
11	CHRISTINA	CASTAGNA		A	ACTIVE	AV	VERIFIED	421 WHITT AVE	BURLINGTON	NC
12	NATHAN	EDWARD		A	ACTIVE	AV	VERIFIED	421 WHITT AVE	BURLINGTON	NC
13	MICHAEL	CHARLES		A	ACTIVE	AV	VERIFIED	107 TERRYWOOD CT	HAW RIVER	NC
14	GENA	HOLT		A	ACTIVE	AV	VERIFIED	107 TERRYWOOD CT	HAW RIVER	NC
15	ALEXANDER						CONFIRMATION NOT RETURNED	1286 PYRTLE DR	GRAHAM	NC
16	MYRA						VERIFIED	612 SIDEVIEW ST	GRAHAM	NC
17	WILLIAM						CONFIRMATION PENDING	3562 S NC HWY 119	HAW RIVER	NC
18	JACK						VERIFIED	612 SIDEVIEW ST	GRAHAM	NC
19	JACK						VERIFIED	612 SIDEVIEW ST	GRAHAM	NC
20	ROBERT						VERIFIED	2220 CRESCENT DR	GRAHAM	NC
21	DEBORAH						VERIFIED	2220 CRESCENT DR	GRAHAM	NC
22	MICHELLE						VERIFIED	201 E CENTER ST #205	MEBANE	NC
23	MICHAEL						VERIFIED	975 SWEET GUM WAY	MEBANE	NC
24	ASHLEY						VERIFIED	975 SWEET GUM WAY	MEBANE	NC
25	MEGAN						VERIFIED	204 WOODALE DR	ELON	NC
26	JOY						VERIFIED	1414 TERRYWOOD RD #307	HAW RIVER	NC
27	RAFAT						MOVED FROM STATE	REMOVED		
28	JENNIFER						VERIFIED	3027 TRUITT DR	BURLINGTON	NC
29	MICHAEL						VERIFIED	3223 BILL LOY DR	ELON	NC
30	TRACY						VERIFIED	504 BROOKFIELD DR	GIBSONVILLE	NC
31	RONALD						VERIFIED	504 BROOKFIELD DR	GIBSONVILLE	NC
32	THOMAS						VERIFIED	250 OBSIDIUM CT	GIBSONVILLE	NC
33	JENNA	CAMILLE		A	ACTIVE	AV	VERIFIED	504 BROOKFIELD DR	GIBSONVILLE	NC
34	ANTHONY	JOSEPH		A	ACTIVE	AV	VERIFIED	504 BROOKFIELD DR	GIBSONVILLE	NC

Value distribution

VOTER_STATUS_REASON_DESC_STATISTICS



Manual profiling example using Excel

	D	E	F	G	H	I	J	K	L	M
		last_name	first_name	middle_name	name_suffix_lbl	status_cd	voter_status_desc	reason_cd	voter_status_reason_desc	res_street_address
311	05378	ADAMS	PAUL	WENDELL				DI	UNAVAILABLE ESSENTIAL INFORMATION	1426 CLOVERDALE ST
622	04302	AGNEW	MATTHEW	PATRICK				DI	UNAVAILABLE ESSENTIAL INFORMATION	615 TRACY DR
664	14513	AGUIAR	ROGELIO	LOPEZ				DI	UNAVAILABLE ESSENTIAL INFORMATION	310 N IRELAND ST
727	17943	AHERN	KATHLEEN	MARY				DI	UNAVAILABLE ESSENTIAL INFORMATION	1720 ST MARK'S CHURCH RD
1224	13639	ALESSANDRO	GUY	THOMAS				DI	UNAVAILABLE ESSENTIAL INFORMATION	600 N MARSHALL ST
1912	07109	ALLISON	LARRY	DONNELL				DI	UNAVAILABLE ESSENTIAL INFORMATION	425 KERNODLE DR
3137	10544	ANDRES	KEITH	CHARLES				DI	UNAVAILABLE ESSENTIAL INFORMATION	108 FLORENCE ST
3341	14169	ANGE	JOSEPH	FENNER	JR			DI	UNAVAILABLE ESSENTIAL INFORMATION	4315 NC HWY 54
3749	13749	ARGENTO	JANIE	CAROL				DI	UNAVAILABLE ESSENTIAL INFORMATION	5533A KIMESVILLE RD
5244	05417	BAILIFF	JAMIE	CATES				DI	UNAVAILABLE ESSENTIAL INFORMATION	1507 TARLETON AVE
5466	12314	BAKER	JOSHUA	KEITH				DI	UNAVAILABLE ESSENTIAL INFORMATION	1400 ROSLYN DR
6010	14173	BANKS	DALE	CHRISTOPHER				DI	UNAVAILABLE ESSENTIAL INFORMATION	206 N FISHER ST
6972	00661	BARTLETT	DAVID	WAYNE				DI	UNAVAILABLE ESSENTIAL INFORMATION	4135 PREACHER HOLMES RD
7116	13139	BASS	KITTY	R				DI	UNAVAILABLE ESSENTIAL INFORMATION	110 EASTGATE DR
7273	5220	BATRES	RICHARD	A				DI	UNAVAILABLE ESSENTIAL INFORMATION	1903 MALONE RD
7282	2616	BATTEN	MICHAEL	LYNN				DI	UNAVAILABLE ESSENTIAL INFORMATION	2404 N CHURCH ST
7623	3562	BEANE	RICKY	EVERETTE				DI	UNAVAILABLE ESSENTIAL INFORMATION	1721 MEBANE OAKS RD
7934	04614	BEEKER	MARY	E				DI	UNAVAILABLE ESSENTIAL INFORMATION	149 CUTLER TR
8041	2628	BELFIELD	GEORGE	WILLIAM	JR			DI	UNAVAILABLE ESSENTIAL INFORMATION	103 ANTIOCH AVE
8189	12351	BELLAMY	MICHAEL	JOEL	MARTIN			DI	UNAVAILABLE ESSENTIAL INFORMATION	453 QUINTAS AVE
8341	04115	BENES	JOEL	MARTIN				DI	UNAVAILABLE ESSENTIAL INFORMATION	219 MOFFITT HALL
8561	3668	BENO	KATHLEEN	M				DI	UNAVAILABLE ESSENTIAL INFORMATION	404 S FLANNER ST
9055	3572	BIALLY	BRIAN	E				DI	UNAVAILABLE ESSENTIAL INFORMATION	3508 GARDEN RD
9419	0281	BISCHOF	JERRY	LEE				DI	UNAVAILABLE ESSENTIAL INFORMATION	417 WESTHAMPTON DR
9558	06730	BLACK	TERESA	ANN		D	DENIED	DI	UNAVAILABLE ESSENTIAL INFORMATION	734 BURLINGTON AVE
9687	01489	BLACKMAN	LISA	REGINA		D	DENIED	DI	UNAVAILABLE ESSENTIAL INFORMATION	645 GUNN ST
9939	02088	BLAIR	JON	STEWART		D	DENIED	DI	UNAVAILABLE ESSENTIAL INFORMATION	847 TROLLINGWOOD-HAWK
9979	14360	BLAKE	KATHY	JEAN		D	DENIED	DI	UNAVAILABLE ESSENTIAL INFORMATION	1528 S MEBANE ST
10031	18109	BLALOCK	ROBERT	WAYNE	JR	D	DENIED	DI	UNAVAILABLE ESSENTIAL INFORMATION	206 N FISHER ST
10463	12374	BODENHEIMER	TRACY	CAPPS		D	DENIED	DI	UNAVAILABLE ESSENTIAL INFORMATION	1116 HILLSIDE DR
10511	7209	BOGER	DERRICK			D	DENIED	DI	UNAVAILABLE ESSENTIAL INFORMATION	306 BROAD ST
11598	03474	BOWENS				D	DENIED	DI	UNAVAILABLE ESSENTIAL INFORMATION	606 S MEBANE ST

Dependencies among columns (if reason_cd = "DI", then voter_status_desc = "DENIED")

Manual profiling example using Excel

Many more interesting questions about the dataset:

- Possible keys?
- Functional dependencies between columns?
- Correlations between columns?
- Frequent patterns in columns?

	X		Y	Z	AA	AB	AC	AD	AE	
1	ode	full_phone_number	confidential_ind	registr_dt	race_code	ethnic_code	party_cd	gender_code	birth_year	age
2			N	10/01/1984	W	NL	UNA	F		1935
3			N	02/23/2018	W	UN	UNA	F		1978
4	27302		N	10/31/2020	W	UN	REP	M		1966
5	27215	3362291110	N	03/26/1996	W	UN	UNA	F		1976
6	27215	2228834	N	08/15/1989	W	NL	UNA	F		1945
7	27340	2027443411	N	03/07/2012	W	UN	DEM	M		1948
8	27215	2529408167	N	06/01/2020	B	NL	DEM	F		1966
9	27215	3362291110	N	10/10/1994	W	UN	UNA	M		1976
10	27215	3362228834	N	12/05/2013	W	UN	UNA	M		1972
11	27215	3362228834	N	12/05/2013	W	HL	UNA	F		1975
12	27215	3362228834	N	06/06/1990	W	NL	REP	M		1944
13	27258	3365789123	N	08/18/1998	W	NL	REP	F		1972
14	27258	3362667615	N	01/19/2006	W	NL	REP	M		1962
15	27253		N	05/06/2020	W	NL	UNA	M		1987
16	27253	3362644972	N	01/16/2009	W	NL	REP	M		1986
17	27253	3366757112	N	06/12/2017	W	UN	UNA	M		1960
18	27253	3365127765	N	12/02/2008	W	UN	REP	F		1951
19	27258		N	08/10/2018	U	UN	REP	U		1959
20	27253	8133102051	N	06/07/2024	W	UN	REP	F		1950
21	27302	9196327914	N	11/01/2024	W	NL	UNA	F		1970
22	27253	8133100652	N	05/09/2023	W	NL	REP	M		1945
23	27302	7049961866	N	08/26/2016	W	NL	UNA	F		1990
24	27302		N	09/08/2016	W	NL	UNA	M		1989
25	27244	9147724005	N	10/04/2024	W	NL	DEM	U		2003
26	27258	4302953995	N	05/17/2023	W	NL	REP	F		1967
27	27215	3365850666	N	10/26/2012	W	NL	DEM	F		1980
28			N	03/30/2000	A	UN	DEM	F		1952
29	27244	4018353485	N	10/11/2024	W	NL	REP	U		2004
30	27249	3366842638	N	04/15/2021	W	NL	REP	M		2003
31	27249	3363398364	N	12/09/2019	W	NL	REP	F		2002
32	27249	3364499029	N	05/14/1996	W	UN	UNA	M		1976
33	27249		N	10/09/2024	W	NL	UNA	U		1979

ncvoter1

Sheet1

Definition of data profiling

- “Data profiling is the process of examining the data available in an existing data source [...] and collecting statistics and information about that data.”

Wikipedia 01/2025

- “Data profiling refers to the activity of creating small but informative summaries of a database.”

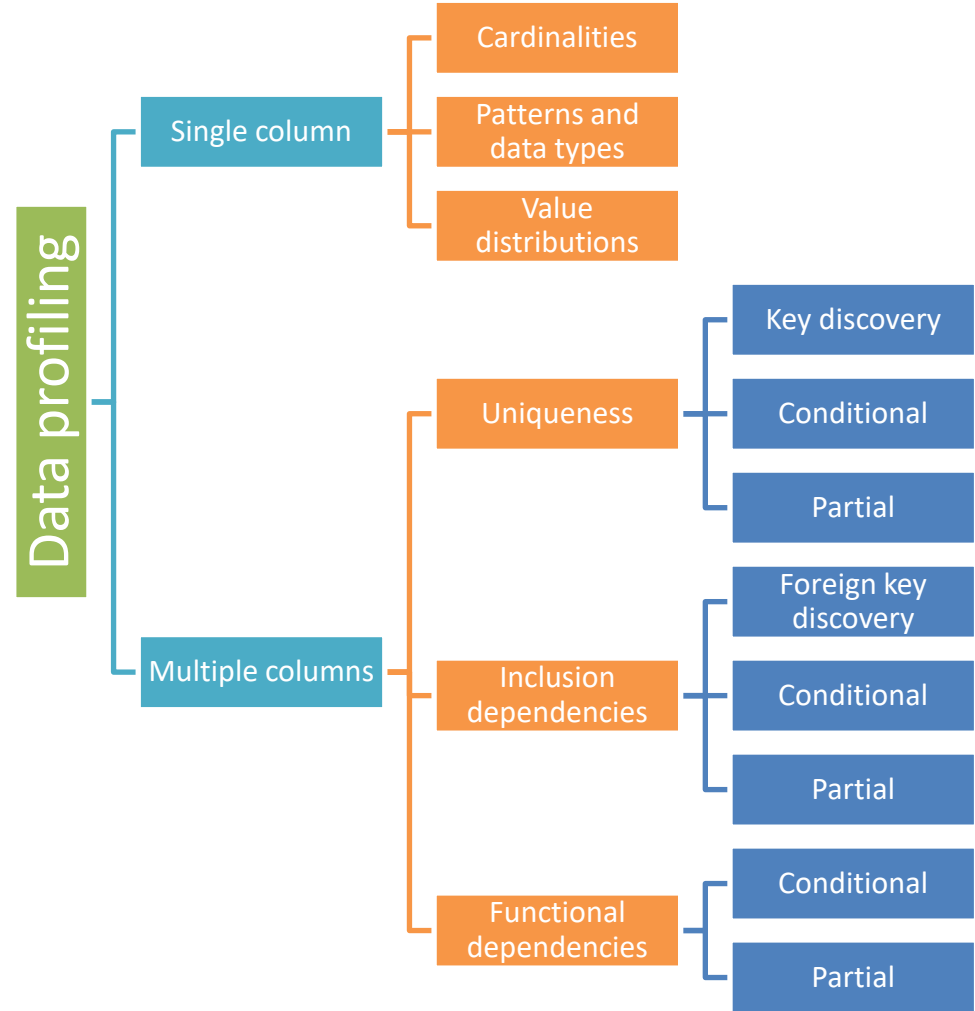
Ted Johnson, Encyclopedia of Database Systems

Data profiling is the set of activities and processes to determine the metadata about a given dataset.

We cover a fixed set of data profiling tasks and their results.

Classification of standard profiling tasks

- Applies to structured (tabular) data.
- Distinguishes between profiling of data stored in a single column and data stored across multiple columns.
- For profiling tasks on multiple columns, exact, conditional, and partial tasks exist.



Single-column vs. multi-column

Single column profiling

- Most basic form of data profiling
- Assumption: All values are of same type
- Assumption: All values have some common properties that are to be discovered
- Often part of the basic statistics gathered by DBMS
- Complexity: Number of values/rows

Multi-column profiling

- Discover joint properties
- Discover dependencies
- Complexity: Number of columns and number of values

Single-column vs. multi-column

Single column profiling

- Most basic form of data profiling
- Assumption: All values are of same type
- Assumption: All values have some common properties that are to be discovered
- Often part of the basic statistics gathered by DBMS
- Complexity: Number of values/rows

Single column

Cardinalities

Patterns and data types

Value distributions

Running example for single column profiling

- Relies on [ydata Profiling](#)
- Integrates into Python Notebooks
- Example exploration of the [Titanic Dataset](#)
- Full ydata Profiling report for this dataset available on the [ydata examples page](#).

PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin
1	0	3	Braund, Mr. Owen Harris	male	22	1	0	A/5 21171	7.25	
2	1	1	Cumings, Mrs. John Bradley (Florence Briggs Thayer)	female	38	1	0	PC 17599	71.2833	C85
3	1	3	Heikkinen, Miss. Laina	female	26	0	0	STON/O2. 3101282	7.925	
4	1	1	Futrelle, Mrs. Jacques Heath (Lily May Peel)	female	35	1	0	113803	53.1	C123
5	0	3	Allen, Mr. William Henry	male	35	0	0	373450	8.05	
6	0	3	Moran, Mr. James	male		0	0	330877	8.4583	
7	0	1	McCarthy, Mr. Timothy J	male	54	0	0	17463	51.8625	E46
8	0	3	Palsson, Master. Gosta Leonard	male	2	3	1	349909	21.075	
9	1	3	Johnson, Mrs. Oscar W (Elisabeth Vilhelmina Berg)	female	27	0	2	347742	11.1333	
10	1	2	Nasser, Mrs. Nicholas (Adele Achem)	female	14	1	0	237736	30.0708	
11	1	3	Sandstrom, Miss. Marguerite Rut	female	4	1	1	PP 9549	16.7	G6
12	1	1	Bonnell, Miss. Elizabeth	female	58	0	0	113783	26.55	C103
13	0	3	Saunderscock, Mr. William Henry	male	20	0	0	A/5. 2151	8.05	

Excerpt of Titanic Dataset

Cardinalities

- Number of values
- Number of distinct values
- Number of NULLs
- MIN, MAX, MEDIAN, AVG value
- **Applications:** data understanding, attribute assessment, attribute categorization, query optimization

Cardinalities

Age

Real number ($\mathbb{R}$)

Age

Missing

Distinct	88
Distinct (%)	12.3%
Missing	177
Missing (%)	19.9%
Infinite	0
Infinite (%)	0.0%
Mean	29.699118

Minimum	0.42
Maximum	80
Zeros	0
Zeros (%)	0.0%
Negative	0
Negative (%)	0.0%
Memory size	7.1 KiB

Cardinalities for the
Age Attribute

Number of distinct values
Number of missing values
MIN, MAX, MEAN Age
...

High number of
missing values
may be a problem
for age-related
analysis.

Data types and value patterns

- String vs. number
- String vs. number vs. date
- Categorical vs. continuous
- SQL data types
`CHAR`, `INT`, `DECIMAL`, `TIMESTAMP`, `BIT`, `CLOB`, ...
- Domains
`VARCHAR(12)` vs. `VARCHAR(13)`
- XML data types (even more fine grained)
- Regular expressions
`(\d{3})-(\d{3})-(\d{4})-(\d+)`
- Semantic domains
Address, phone, email, first name
- **Applications:** data understanding, data modeling,
attribute categorization, query optimization



Increasing difficulty

Data types and value patterns

Variable types impact how the attributes are to be handled prior / during analysis.

Summary of detected types

Age
Real number ($\mathbb{R}$)
Missing

Embarked
Categorical
Distinct

Dataset statistics

Number of variables	12
Number of observations	891
Missing cells	866
Missing cells (%)	8.1%

Variable types

Numeric	5
Categorical	4
Text	3

Distinct

Distinct (%)

Missing

Missing (%)

Infinite

Infinite (%)

Mean

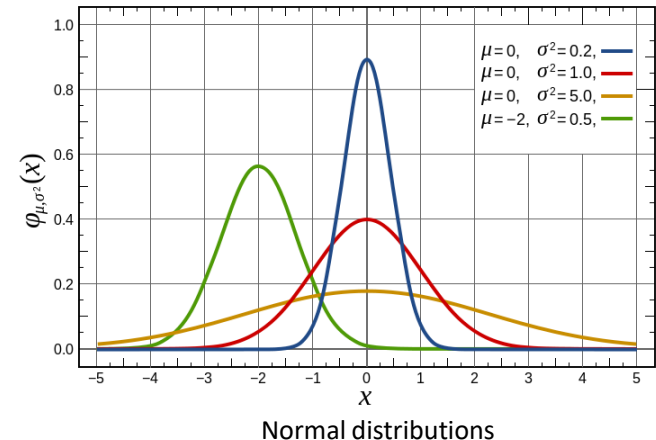
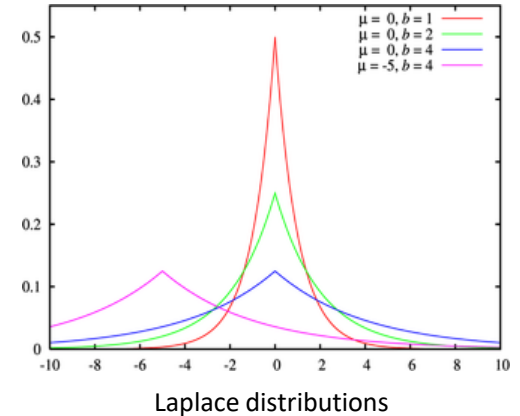
Overview **Alerts** 10 **Reproduction**

Inferred from CSV

Inferred from CSV

Numeric distributions

- Probability distribution for numeric values
- Detect whether data follows some well-known distribution
 - Determine that distribution function for data values
- If no specific/useful function detectable: histograms



Histograms

- Determine (and display) value frequencies for value intervals or for individual values
- Estimation of probability distribution for continuous variables
- Applications: Query optimization, outlier detection, visualization

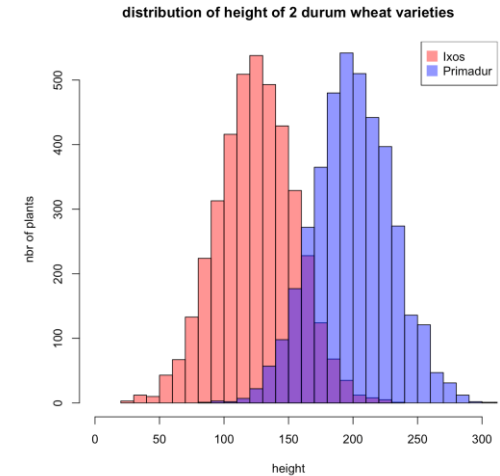


Image source: [R Graph Gallery](#)

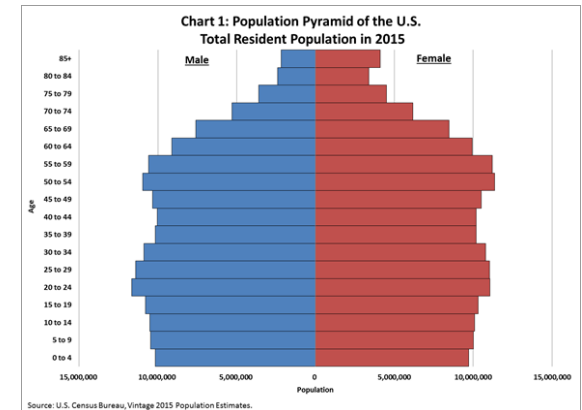
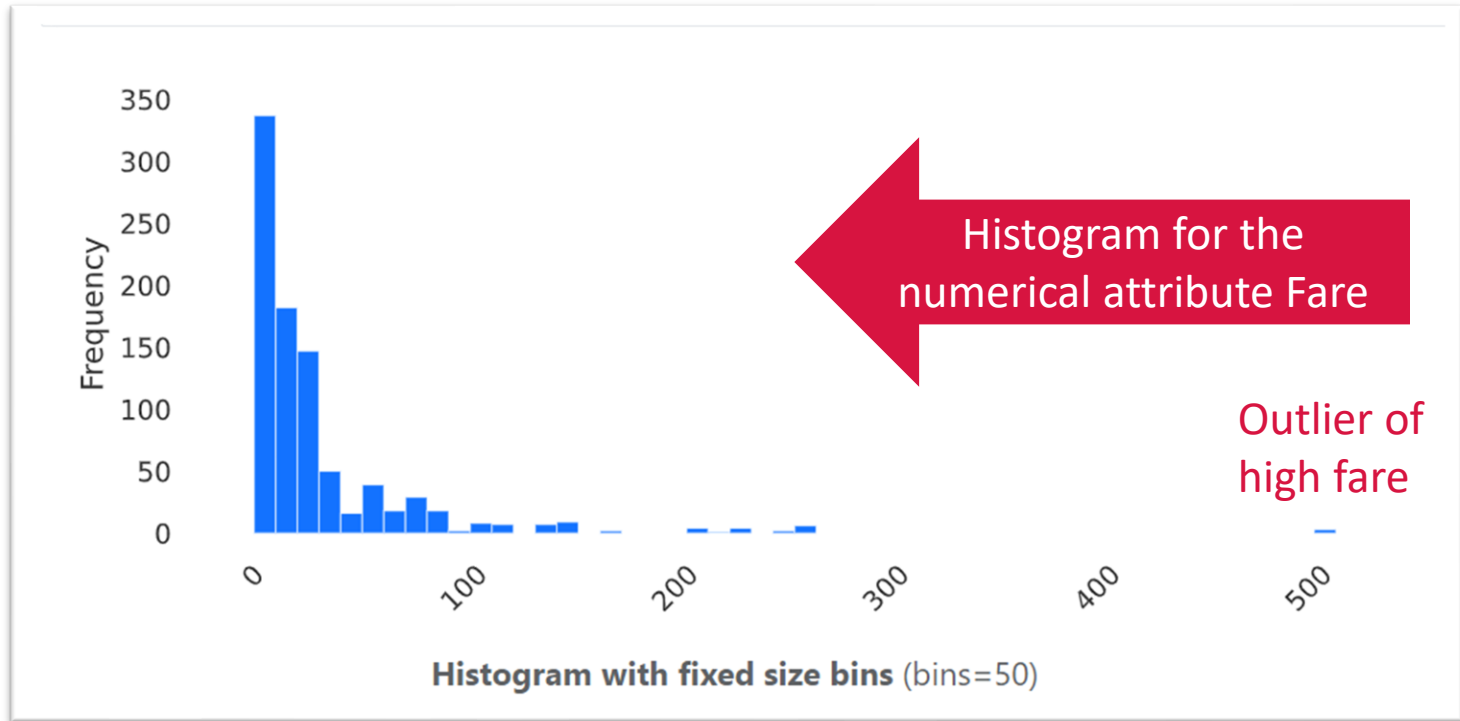


Image source: [US Census](#)

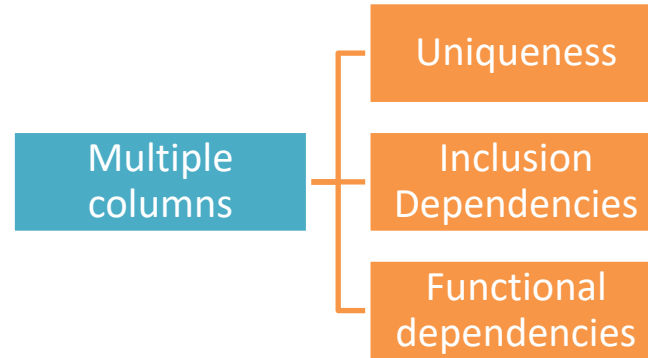
Histograms



Computational complexity

- All single-column profiling tasks require to scan the data of a table once.
- While reading in each row, variables or data structures are updated, from which the final result is derived.
- Computational dependency depends on the number of values / rows.
- This is considered easy :)

Next week preview:



Multi-column profiling

- Discover joint properties
- Discover dependencies
- Complexity: Number of columns and number of values

References and credits

- Joseph M. Hellerstein, Tye Rattenbury, Jeffrey Heer, Sean Kandel, Connor Carreras. Principles of Data Wrangling. O'Reilly Media. 2017 – Chapter 4
- Ziawasch Abedjan, Lukasz Golab, Felix Naumann: Profiling relational data: a survey. VLDB J. 24(4): 557-581 (2015)
- ***Slides partially based on slides shared by Felix Naumann, Hasso-Plattner-Institut Potsdam, Germany***